

Taoyuan Wang

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EXPERIENCE

Software Engineer at Wing (Alphabet/ex-Google)

Oct 2022 – Present

UAS (Uncrewed Aircraft Systems) Traffic Management (UTM) Team

Palo Alto, CA

- Designed and developed distributed **real-time** applications for Google X's autonomous drone delivery initiative, advancing next-generation aviation systems with safety validated through automated end-to-end testing
- Engineered **C++/Scaffolding** backend services using **Protocol Buffers** and **S2 Geometry Library** to optimize geofencing and geospatial clustering
- Architected telemetry pipelines on Alphabet infrastructure for historical fleet visualization, utilizing **Google Cloud Pub/Sub** and **Flume**-based GeoStore importers for metric monitoring and alerting, integrated with **Spanner** Change Data Capture (CDC) and **gRPC**-Web client streaming for UI updates
- Built **Google Maps** operator interfaces for 4D (3D + time) airspace management, leveraging **Gemini** to generate airspace restrictions from real-world events to reduce operator cognitive load and response time
- Modernized authentication with role-based access control and **low-latency** authorization for safety-critical operations, including pilot-initiated emergency landing
- Led development of a proof-of-concept **LLM/VLA**-powered multi-drone coordination system featuring agentic planning with iterative review, human-in-the-loop escalation, and safety validation, enabling natural language control, AI-human-robot collaboration, and **3D simulation** for rapid scenario generation

Software Engineer at Appen (fka Figure Eight)

Oct 2021 – Oct 2022

Computer Vision Tools Team (Internal transfer from China)

Sunnyvale, CA

- Led end-to-end development of AI-assisted annotation tools for **multi-modal** data (React/Three.js frontend and Flask/Spring backend), from prototype to production
- Extended point cloud cuboid annotation to **semantic segmentation** as a side project, later adopted as a new product used by 20+ leading clients in autonomous vehicles and robotics sectors

Software Engineer at Appen China

Apr 2019 – Sep 2021

Computer Vision Tools Team

Shanghai, China

- Developed a localized platform for the Chinese market as the founding engineer at Appen China
- Explored point cloud **object detection/tracking** to enhance labeling efficiency using open-source projects
- Conducted **sensor fusion** experiments using NumPy to project 3D LiDAR points into 2D camera views based on calibration parameters (e.g., position, rotation, focal length) from client devices
- Implemented event loggers, collaborating with UX designers and data scientists to analyze annotator behavior for training process and tools improvement, resulting in a 56% reduction in labeling costs

Software Engineer at Works Applications

Apr 2018 – Mar 2019

AI WORKS HUE Tech Lead Group

Shanghai, China

- Built full-stack ERP solutions with Vue.js frontend and Spring/Hibernate backend
- Explored **transformer**-based chatbot and LSTM-based OCR (Tesseract) for workflows automation

Research Assistant at Academia Sinica

Dec 2017 – Mar 2018

Natural Language Processing Lab

Taipei, Taiwan

- Researched conversational agents for Chinese short text and maintained large Weibo post-comment dataset
- Implemented **Seq2Seq** with **attention** in PyTorch and applied **reinforcement learning** from human feedback (RLHF) using comment appropriateness scores as rewards

IT Specialist at Social Affairs Bureau (Military Service)

Nov 2016 – Oct 2017

EDUCATION

National Tsing Hua University

Hsinchu, Taiwan

M.S. in Computer Science (Graduate Researcher at Internet Lab)

Sep 2015 – Aug 2016

B.S. in Computer Science (Exchanged to Beijing Tsinghua and Nagoya University)

Sep 2010 – Aug 2015

SKILLS

Languages: C/C++, Python, Java, JavaScript/TypeScript

Frameworks & Libraries: Flask, pytest, Spring, JUnit, React/Redux, Jest, Three.js (WebGL), NumPy

Tools & Infras: Git, Docker, Jenkins, Jetski (aka Google Antigravity), GCP, AWS